CARTESIAN OBJECTS OF D(O) FROM CARTESIAN SIMPLICIAL SYSTEMS OF MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The label lemma-cartesian-module-derived-from-simplicial names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The counted proof is the author's complete argument: a parallel-proof delegation to the registered sibling lemma-cartesian-object-derived-from-simplicial (card X1-sp-simplicial-cartesian-object-derived-from-simplicial) whose five changes are each explicitly specified in the source text - accepted per the good-sub / criterion-very-ample-bis sister-construction precedents.

Allowed standard prerequisites (editorial).

- Sibling: cartesian object of D(Z) from a simplicial system. Same chapter, \label{lemma-cartesian-object-derived-from-simplicial} (registered card X1-sp-simplicial-cartesian-object-derived-from-simplicial). Line 3556: the counted proof is the sibling's with the five listed changes.
- Simplicial resolution over a ringed site. Same chapter, \label{lemma-simplicial-resolution-ringed}. Change (3) at line 3562: replaces lemma-simplicial-resolution-Z-site.
- Postnikov comparison for modules. Same chapter, \label{lemma-modules-postnikov}. Change (5) at line 3567: replaces the abelian Postnikov comparison.

Source reference. Original lemma 0D9L: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3175–3175. Prerequisite pointer invoked at proof line(s) 3556; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D9B: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2092–2092. Prerequisite pointer invoked at proof line(s) 3562; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0D9Q: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3448–3448. Prerequisite pointer invoked at proof line(s) 3567; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D9I: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3002–3002. Prerequisite pointer invoked at proof line(s) 3567; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at statement line(s); it names the Situation supplying the running notation of the chapter, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09WI: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1711–1711. Prerequisite pointer invoked at proof line(s) 3562; it names the standard fact that the counted argument's change (3) explicitly replaces, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$. Let (K_n, K_φ) be a simplicial system of the derived category of modules. Assume*

(1) *$f_\varphi^{-1}\mathcal{O}_n \to \mathcal{O}_m$ is flat for $\varphi : [m] \to [n]$,*
(2) *(K_n, K_φ) is cartesian,*
(3) *$\mathrm{Hom}(K_i[t], K_i) = 0$ for $i \geq 0$ and $t > 0$.*

Then there exists a cartesian object K of $D(\mathcal{O})$ whose associated simplicial system is isomorphic to (K_n, K_φ).

Proof. The proof is exactly the same as the proof of Lemma 0D9L with the following changes

(1) use $g_n^* = Lg_n^*$ everywhere instead of g_n^{-1},
(2) use $f_\varphi^* = Lf_\varphi^*$ everywhere instead of f_φ^{-1},
(3) refer to Lemma 0D9B instead of Lemma 09WI,
(4) in the construction of $Y'_{m,n}$ use $\mathcal{O}_m$ instead of $\mathbf{Z}$,
(5) compare with the proof of Lemma 0D9Q rather than the proof of Lemma 0D9I.

This ends the proof. □